

Project Proposal - Machine Learning (MITS 6800G)

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Title: IoT Botnet Detection via Network Traffic Analysis Using Machine Learning

Problem Statement

The growth of Internet of Things (IoT) devices has led to the rise of security issues related to the limited processing capabilities of the devices, inadequate authentication, and poor patch management. These issues have made IoT devices prone to botnets such as Mirai and Bashlite, which are used in carrying out distributed denial of service (DDoS) attacks, data theft, and reconnaissance. Signature-based intrusion detection systems have difficulty in keeping up with the changing behaviors of botnets, and the high volume of IoT traffic is difficult to analyze manually. The objective of the current project is the development of a machine learning framework that can effectively distinguish between normal and malicious traffic from IoT devices, with the use of the N-BaIoT dataset from the UCI Machine Learning Repository.

Proposed Solutions

This project suggests a machine-learning system to identify IoT botnets based on the analysis of network-traffic by preprocessing raw device traffic, identifying important behavioral patterns, and modeling normal versus malicious network-traffic. The system will analyze indicators, including but not limited to packet statistics, communication flows, distribution-based patterns using both supervised and unsupervised methods in order to classify traffic and provide scores on the frequency of risks as indicators that signal suspicious activities. Developed and tested on the N-BaIoT dataset, the framework will offer a flexible, data-driven platform that will be capable of detecting the changing attacks of the botnets and reinforce the security of the IoT networks.

Machine Learning Concepts that will be implemented in this project:

- Extracting features of flow-based network data
- Statistical behavior Modeling of network traffic
- Learning techniques (Random Forest, Support Vector Machines, Logistic Regression, Neural Networks) which are supervised
- Precision, recall, F1 score and ROC AUC Model evaluation
- Hyperparameter tuning Cross-validation

Datasets

- IoT Botnet Detection Dataset (N-BaIoT) –
<https://archive.ics.uci.edu/dataset/442/detection+of+iot+botnet+attacks+n+baiot>
- Bot-IoT Dataset (UNSW Canberra) –
https://unsw-my.sharepoint.com/:f:/g/personal/z5131399_ad_unsw_edu_au/EjlBDf2KODxPgXmqbO3MxxsBBVARCKZxGUG47OiFHb_AnQ
- CICIDS2017 Network Intrusion Dataset –
<https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset>

Reference Papers

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